COMSATS University Islamabad, Lahore Campus

Mid Term, Examination – Spring 2024

Course Title:	Web Technologies			Course Code:	CSC336	Credit:	4(3,1)
Course Instructor/s:	Muhammad Usman Akram			Programme Name:	BSCS		
Semester:	6	Batch:	FA21	Section:	A,B	Date:	Apr. 24, 2024
Time Allowed:	1.5 Hour			Maximum Marks:	50		
Student's Name:				Reg. No:			

Important Instructions / Guidelines:

Please implement functionality that is asked, do not do any extra work. Present your work with good UI.

Question 1 [CLO-1] **[5*3 = 15 marks]**

1. Explain the different parts of the CSS box model (content, padding, border, margin) and how they contribute to the overall dimensions and positioning of an element on the web page.
2. What is the Document Object Model (DOM) in web development? How does JavaScript interact with the DOM?
3. Explain with an example how a callback based function achieves asynchronous behavior in JS

Question 2 [CLO-2] **[10 marks]**

(Do not use Bootstrap for this) Using media queries and mobile first design principle, write HTML and CSS such that, there shall be 4 links that shall appear vertically stacked on small devices but for devices with 400px or higher width, links shall appear horizontal i.e. in a single row (assume they are header links). [10 Marks]

Question 3 [CLO-2] **[15 marks]**

Consider the following html code, assume the page already contains jquery.

```
<div id="book-library">
   <input type="text" id="title" placeholder="Title">
   <input type="text" id="author" placeholder="Author">
   <button id="add-book">Add Book</button>
   <ul id="books">
      <!-- Books will be added here -->
   </ul>
</div>
```

Write jQuery code to implement CRUD operations for books:

- Create: Add a new book to the library when the "Add Book" button is clicked.
- Read: Display all books from an array when the page loads.
- Update: Allow updating a book's details by clicking on it.
- Delete: Remove a book when delete button is clicked.

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Owner of SuperCUI

Question 4 [CLO-3] **[10 marks]**

Assume an HTTP end point *example.com/products.json* returns JSON i.e. an array of objects where each object contains different attributes of a product. Write only JavaScript code, such that, on page load, it shall fetch products using AJAX and create an HTML table. Embed the created table in inner HTML of an already existing div with id "canvas". You must display all attributes of each product, so the number of columns shall be equal to number of product attributes. Assume all products have same field names and all fields are initialized with an atomic value of type number, string, Boolean, etc, (but not an array).



CUI
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